

A Data Driven Resolvent Framework for Generalized Transfer Operators

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1 Background

1.1 One Parameter Semigroups

The theory of semigroups provides a robust framework for studying solutions to linear, constant ordinary differential equations on Banach spaces.

Let T be a family of operators

$$T(h) : X \rightarrow X, \quad h \geq 0,$$

where X is a Banach space. The family $\{T(h)\}_{h \geq 0}$ is called a *one-parameter semigroup* if it satisfies:

1. **Semigroup Property (Associativity):**

$$T(h + s) = T(h)T(s), \quad \forall h, s \geq 0.$$

2. **Identity:**

$$T(0) = I,$$

where I denotes the identity operator on X .

Strong Continuity. A one-parameter semigroup $\{T(h)\}_{h \geq 0}$ is said to be *strongly continuous* (or a C_0 -semigroup) if for every fixed $x \in X$,

$$\lim_{h \rightarrow 0^+} \|T(h)x - x\| = 0.$$

This means that for each $x \in X$, the mapping $h \mapsto T(h)x$ is continuous with respect to the norm on X .

Uniform Continuity. In contrast, the semigroup is called *uniformly continuous* if the mapping $h \mapsto T(h)$ is continuous in the operator norm. That is,

$$\lim_{h \rightarrow 0^+} \|T(h) - I\| = 0.$$

Uniform continuity is a stronger condition than strong continuity. Indeed, if $\{T(h)\}$ is uniformly continuous, then

$$\lim_{h \rightarrow 0^+} \|T(h)x - x\| \leq \lim_{h \rightarrow 0^+} \|T(h) - I\| \|x\| = 0,$$

for all $x \in X$, but the converse need not hold.

Infinitesimal Generator. For a strongly continuous semigroup, the *infinitesimal generator* A is defined by

$$Ax = \lim_{h \rightarrow 0^+} \frac{T(h)x - x}{h},$$

for all x in the domain

$$D(A) = \left\{ x \in X : \lim_{h \rightarrow 0^+} \frac{T(h)x - x}{h} \text{ exists} \right\}.$$

If the semigroup is uniformly continuous, then A is a bounded linear operator, and the semigroup can be represented by the exponential series:

$$T(h) = e^{hA} = \sum_{k=0}^{\infty} \frac{(hA)^k}{k!}.$$

This representation is valid because uniform continuity guarantees the convergence of the series in the operator norm.

1.2 Local Linearizations

Consider the Cauchy problem

$$\begin{aligned}\frac{du}{dt} + A(u(t)) &= 0, \quad t > 0, \\ u(0) &= u_0, \quad u_0 \in U \subset X,\end{aligned}\tag{1}$$

where A is an operator (possibly nonlinear and state-dependent) that generates a contraction semi-group $T(h)$ on the Banach space X . The contraction property aids in establishing the existence and uniqueness of solutions, yet it also means that a global linearization of A may be unattainable when its dependence on $u(t)$ is significant.

In many practical settings, one only has access to discrete observations of the state. Let these observations be arranged in the data matrix

$$U_t = \begin{bmatrix} u(t) \\ u(t-h) \\ u(t-2h) \\ \vdots \end{bmatrix},$$

where the rows correspond to states sampled at time intervals of length h .

A first approach is to attempt a global linearization via a least-squares formulation:

$$U_t \approx U_{t-h} T(h),\tag{2}$$

$$\text{or equivalently, } u(t) \approx T(h) u(t-h).\tag{3}$$

If the residuals from this approximation (measured in an appropriate norm) are sufficiently small—i.e., within an acceptable and prescribed tolerance—then it is reasonable to assume that the dynamics are well represented by a linear operator. In such a case, one can recover the global generator via

$$A = \lim_{h \rightarrow 0^+} \frac{1}{h} (T(h) - I).$$

This approach is particularly effective when A is linear or nearly linear.

However, when A is strongly nonlinear or state-dependent, a global linearization may fail to capture the system's behavior. In these situations, a local linearization is preferable. To construct a local approximation, we introduce a weighting operator W designed to select or emphasize those rows in U_t that are dynamically relevant. The idea is to filter the data so that the local dynamics around the state $u(t)$ are highlighted. The weighted observations are given by

$$WU_t,$$

and the local approximation is then expressed as

$$WU_t \approx WU_{t-h} \hat{T}(h, t),\tag{4}$$

where $\hat{T}(h, t)$ is the locally linearized evolution operator valid in a neighborhood of $u(t)$.

The quality of the local linearization can be assessed by examining the residuals of the least-squares fit in (4). If the residual error exceeds a predetermined threshold, it may indicate that even the local approximation is insufficient, prompting further refinement or alternative modeling strategies.

In the limit as $h \rightarrow 0$, the local infinitesimal generator $A(u(t))$ is recovered by

$$A(u(t)) = \lim_{h \rightarrow 0} \frac{1}{h} \left(\hat{T}(h, t) - I \right).$$

Alternatively, if a weighted pseudoinverse is employed, one may write

$$A(u(t)) = \lim_{h \rightarrow 0} \frac{1}{h} \left((WU_{t-h})^+ (WU_t) - I \right),$$

where $(WU_{t-h})^+$ denotes the pseudoinverse of WU_{t-h} .

The selection of the weighting operator W is crucial. In practice, W should be designed to:

- Emphasize regions of the state space where the dynamics are most sensitive.
- Attenuate noise or less informative observations.
- Reflect any prior knowledge about the structure or local behavior of A .

This local linearization strategy thus provides a framework for approximating the state-dependent generator $A(u(t))$ from data, even in the presence of significant nonlinearities that preclude a global linear model.

2 Nonlocality in Infinitesimal Generators

A system's evolution operator may be classified as *local* or *nonlocal* in both space and time. This gives rise to four possible cases:

- **Local in Space, Local in Time:** Standard differential operators such as the heat equation, $A = \partial_t + \Delta_x$.
- **Nonlocal in Space, Local in Time:** Fractional Laplacians and spatial integral operators:

$$Af(x) = \int_{\Omega} K(x, y)(f(y) - f(x))dy. \quad (5)$$

- **Local in Space, Nonlocal in Time:** Memory-dependent PDEs where the evolution depends on past states:

$$\frac{\partial f}{\partial t} = \int_0^t K(t, s) \Delta_x f(s, x) ds \quad (6)$$

- **Nonlocal in Space, Nonlocal in Time:** Systems with both spatial and temporal integral operators:

$$\frac{\partial f}{\partial t} = \int_0^t \int_{\Omega} K(t, s, x, y) f(s, y) dy ds \quad (7)$$

2.1 Nonlocal Semigroups and Generators

Consider the integral operator

$$(T(h)f)(\xi) = \int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta \quad (8)$$

with infinitesimal generator

$$(Af)(\xi) = \lim_{h \rightarrow 0} \frac{1}{h} (T(h) - I) f(\xi) \quad (9)$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} ((T(h)f)(\xi) - f(\xi)) \quad (10)$$

By substitution

$$(Af)(\xi) = \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta - f(\xi) \right) \quad (11)$$

The goal is to rewrite the integral term so that $f(\xi)$ appears inside the integral. We add and subtract $K_h(\xi, \eta) f(\xi)$ inside the integral,

$$(Af)(\xi) = \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} K_h(\xi, \eta) f(\eta) - K_h(\xi, \eta) f(\xi) + K_h(\xi, \eta) f(\xi) d\eta - f(\xi) \right) \quad (12)$$

and factor,

$$(Af)(\xi) = \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} K_h(\xi, \eta) (f(\eta) - f(\xi)) d\eta + \int_{\Omega} K_h(\xi, \eta) f(\xi) d\eta - f(\xi) \right) \quad (13)$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \int_{\Omega} K_h(\xi, \eta) (f(\eta) - f(\xi)) d\eta + \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} K_h(\xi, \eta) d\eta - 1 \right) f(\xi) \quad (14)$$

We can ensure that $K_h(\xi, \eta)$ acts as a state transition probability density by imposing the normalization condition,

$$\int_{\Omega} K_h(\xi, \eta) d\eta = 1 \quad (15)$$

while at the same time eliminating the second term,

$$\lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\Omega} K_h(\xi, \eta) d\eta - 1 \right) f(\xi) = 0 \quad (16)$$

thus recovering the infinitesimal generator as a standard non-local operator,

$$(Af)(\xi) = \lim_{h \rightarrow 0} \frac{1}{h} \int_{\Omega} K_h(\xi, \eta) (f(\eta) - f(\xi)) d\eta \quad (17)$$

Kernel Structure: The resolvent operator associated with A is given by:

$$R_\lambda(A) = (\lambda I - A)^{-1}. \quad (18)$$

It is well known (see, e.g. Pazy) that the semigroup $T(h)$ can be expressed in terms of the resolvent via the contour integral,

$$T(h) = \frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} R_\lambda(A) d\lambda \quad (19)$$

We equate the integral representations

$$\int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta = \frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} R_\lambda(A) f(\xi) d\lambda \quad (20)$$

Furthermore, Arendt & Bukhvalov showed that if $T(h)$ is an integral operator, then so is the resolvent,

$$(R_\lambda f)(\xi) = \int_{\Omega} M_\lambda(\xi, \eta) f(\eta) d\eta \quad (21)$$

where $M_\lambda(\xi, \eta)$ is the resolvent kernel. From (20) using the integral resolvent,

$$\int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta = \frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} \int_{\Omega} M_\lambda(\xi, \eta) f(\eta) d\eta d\lambda \quad (22)$$

Two cases follow regarding the absolute integrability of

$$F(\lambda, \eta) = e^{h\lambda} M_\lambda(\xi, \eta) f(\eta)$$

Case 1

If $F \in L^1(\Gamma \times \Omega)$ then for

$$\oint_{\Gamma} \int_{\Omega} |F(\lambda, \eta)| d\mu(\eta) d\nu(\lambda) \quad (23)$$

we may apply Fubini's theorem and interchange the order of integration

$$\int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta = \int_{\Omega} \left(\frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} M_\lambda(\xi, \eta) d\lambda \right) f(\eta) d\eta \quad (24)$$

Since this must hold for all f , we identify the kernels,

$$K_h(\xi, \eta) = \frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} M_\lambda(\xi, \eta) d\lambda \quad (25)$$

Since $T(0) = I$, the kernel at $h = 0$ must satisfy

$$K_0(\xi, \eta) = \delta(\xi - \eta) \quad (26)$$

In many classical settings, one can show that the resolvent satisfies

$$(\lambda I - A)R_\lambda(A) = I \quad (27)$$

so that in kernel form

$$(\lambda I - A)M_\lambda(\xi, \eta) = \delta(\xi - \eta) \quad (28)$$

$M_\lambda(\xi, \eta)$ serves as the Green's function for the operator $\lambda I - A$. Supposing that for values of λ in a suitable region the Neumann series expansion hold

$$(\lambda I - A)^{-1} = \sum_{n=0}^{\infty} \frac{A^n}{\lambda^{n+1}} \quad (29)$$

Then the kernel $M_\lambda(\xi, \eta)$ may be expressed as

$$M_\lambda(\xi, \eta) = \sum_{n=0}^{\infty} \frac{A^n}{\lambda^{n+1}} \delta(\xi - \eta) \quad (30)$$

$$M_\lambda(\xi, \eta) = \sum_{n=0}^{\infty} \frac{A^n}{\lambda^{n+1}} \delta(\xi - \eta) \quad (31)$$

Substitute this expansion into

$$K_h(\xi, \eta) = \frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} \left(\sum_{n=0}^{\infty} \frac{A^n}{\lambda^{n+1}} \delta(\xi - \eta) \right) d\nu(\lambda) \quad (32)$$

$$= \delta(\xi - \eta) \sum_{n=0}^{\infty} A^n \left[\frac{1}{2\pi i} \oint_{\Gamma} \frac{e^{h\lambda}}{\lambda^{n+1}} d\nu(\lambda) \right] \quad (33)$$

According to the residue theorem

$$\frac{1}{2\pi i} \oint_{\Gamma} \frac{e^{h\lambda}}{\lambda^{n+1}} d\nu(\lambda) = \frac{h^n}{n!} \quad (34)$$

Substituting this back into (33), we obtain

$$K_h(\xi, \eta) = \delta(\xi - \eta) \sum_{n=0}^{\infty} \frac{h^n}{n!} A^n = \delta(\xi - \eta) e^{hA} \quad (35)$$

Thus,

$$K_h(\xi, \eta) = \delta(\xi - \eta) e^{hA} \quad (36)$$

which is equivalent to the operator identity,

$$T(h) = e^{hA} \quad (37)$$

In summary, starting from the contour-integral representation of the kernel and the semigroup property $K_0(\xi, \eta) = \delta(\xi - \eta)$, we have derived—via the resolvent expansion and the evaluation of the contour integrals—that the semigroup kernel is given by the exponential of the generator acting on the delta function.

Case 2

Even if

$$\oint_{\Gamma} \int_{\Omega} |F(\lambda, \eta)| d\eta |d\lambda| = \infty \quad (38)$$

the integrals may still be interchanged provided, in addition to the assumed σ -finiteness of $\mu(\eta)$, $\nu(\lambda)$ is also σ -finite. This guarantees the existence of a non-decreasing sequence of non-negative simple functions $\{\psi_n\}$ converging upward to F^+ even where $\int F d(\mu \times \eta) = \infty$. This allows the interchange of integrals by application *Tonelli's theorem*, and we may again identify:

$$K_h(\xi, \eta) = \frac{1}{2\pi i} \oint_{\Gamma} e^{h\lambda} M_{\lambda}(\xi, \eta) d\lambda \quad (39)$$

2.2 The Evolution Kernel

In the classical theory of C_0 semigroups, the evolution operator $T(h)$ composes a C_0 semigroup generated by a linear operator A

$$T(h) = e^{hA}.$$

In this setting, the kernel $K_h(\xi, \eta)$ is the Dirac delta

$$K_h(\xi, \eta) \xrightarrow{h \rightarrow 0} \delta(\xi - \eta).$$

This convergence ensures that, as $h \rightarrow 0$, the evolution operator reproduces the identity operator.

The Nonlinear (State-Dependent) Case: When the system is nonlinear, the evolution is typically given by a state-dependent mapping,

$$T(h)(\xi) = \Phi_h(\xi),$$

we may still maintain an integral representation:

$$(T(h)f)(\xi) = \int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta.$$

In order for the kernel to capture the correct local behavior of the flow, we require that for small h ,

$$K_h(\xi, \eta) \rightarrow \delta(\Phi_h(\xi) - \Phi_h(\eta)).$$

Since Φ_0 is the identity mapping, this condition reduces to

$$\lim_{h \rightarrow 0} K_h(\xi, \eta) = \delta(\xi - \eta),$$

which, in the weak sense, means that for any test function f ,

$$\lim_{h \rightarrow 0} \int_{\Omega} K_h(\xi, \eta) f(\eta) d\eta = f(\xi).$$

3 Transition Kernel Approximations

In many practical applications, the transition kernel $K_h(\xi, \eta)$ of a semigroup may not be directly available. Instead, we rely on empirical observations of the state transitions of the system over small time increments. We can use these observations to form a probabilistic description of system transitions that respects uncertainty from measurement functions or features intrinsic to the system. Two general categories of such transitions have been studied.

In a broad sense we would like to consider transitions as the *pullback* and *pushforward* of a suitable measure on the system's domain and co-domain. Traditionally, for dynamical systems, these have respectively found expression in the *composition* and *transfer operators*. In applied settings the pullback-composition is better known as the *Koopman operator* and the pushforward-transfer as the *Perron-Frobenius* or *Ruelle-Perron-Frobenius operator*.

We use Local Globally Weighted Least Squares (LGWLS) to construct approximations of a system's Markov transition kernel. This construction is then leveraged to provide an approximation of the stochastic generalizations of the *Perron-Frobenius* and *Koopman operators*.

A quick reminder on notation; it's helpful to remember that, for dynamical systems, its tradition to express the system's evolution as $x_{t+1} = f(x_t)$. This defines naturally a forward map that describes trajectories:

$$x_0 \rightarrow x_1 = f(x_0) \rightarrow x_2 = f(x_1) \rightarrow \dots \quad (40)$$

Consequently, $x = f(y)$ naturally means “if we are at y now, we will be at x in the future”.

3.1 Generalized Perron-Frobenius and Koopman Operators

Let X and Y be spaces equipped with measures μ and ν respectively.

Consider the map:

$$f : Y \rightarrow X \quad (41)$$

For all measurable sets $A \subseteq X$, the *pushforward measure* of f is,

$$f_*(\mu)(A) = \mu(f^{-1}(A)) \quad (42)$$

For all measurable sets $B \subseteq Y$, the *pullback measure* of f is,

$$f_b(\nu)(B) = \nu(f(B)) \quad (43)$$

Koopman and Perron-Frobenius Operators

The Koopman operator, U , evolves observables of X , $g \in L^2(X, \mu)$,

$$(Ug)(x) = g(f(x)) \quad (44)$$

The Perron-Frobenius operator P evolves densities, $d \in L^1(X, \mu)$,

$$(Pd)(x) = \int_{y \in f^{-1}(x)} k(x, y) d(y) dy \quad (45)$$

where k is a transition kernel. For a deterministic system the kernel is the Dirac delta distribution

$$k(x, y) = \delta(x - f(y)) \quad (46)$$

If f is smooth and invertible then a change of variables yields the standard form of P for deterministic f ,

$$(Pd)(x) = \int_X \delta(x - f(y)) d(y) dy \quad (47)$$

$$= \sum_{y \in f^{-1}(x)} \frac{d(y)}{|Df(y)|} \quad (48)$$

The denominator, $|Df(y)|$, accounts for local expansion or contraction under f .

Duality of U and P

The Koopman and Perron-Frobenius operators form an adjoint pair under the standard inner product,

$$\langle Pd, g \rangle = \langle d, Kg \rangle \quad (49)$$

One relatively straightforward way to contextualize this fact is to assume that μ is a Borel probability measure. In this case, the inner products,

$$\langle Pd, g \rangle_\mu = \langle d, Kg \rangle_\mu \quad (50)$$

provide expectation values based on the evolution of f .

Consider,

$$\langle Pd, g \rangle_\mu = \int_X (Pd)(x) g(x) d\mu(x) \quad (51)$$

$$= \int_X \left(\int_X k(x|y) d(y) d\mu(y) \right) g(x) d\mu(x) \quad (52)$$

$$(53)$$

To reiterate:

- $d(y)$ is the initial state probability density on X
- k is a Markov transition kernel providing, given y , the likelihood the next state is x
- $g(x)$ is the observation defined at x

We have,

$$\langle Pd, g \rangle_\mu = \int_X g(x) \left(\int_X k(x|y) d(y) d\mu(y) \right) d\mu(x) \quad (54)$$

$$= \int_X d(y) \left(\int_X k(x|y) g(x) d\mu(x) \right) d\mu(y) \quad (55)$$

If f is deterministic then $k = \delta(x - f(y))$ sifts $\int_X k(x|y) g(x) d\mu(x) = g(f(y))$, to yield

$$= \int_X d(y) \left(\int_X \delta(x - f(y)) g(x) d\mu(x) \right) d\mu(y) \quad (56)$$

$$= \int_X d(y) g(f(y)) d\mu(y) \quad (57)$$

which is just $\mathbb{E}_{y \sim d} [g \circ f(y)]$. If f is not deterministic and k is a general Markov kernel, $\int_X k(x|y) g(x) d\mu(x)$ is the expected value of g given the previous state y . Either way,

$$= \int_X d(y) \left(\int_X k(x|y) g(x) d\mu(x) \right) d\mu(y) \quad (58)$$

$$= \mathbb{E}_{Y \sim d} [g(f(Y))] \quad (59)$$

which is just the extension of the Koopman operator for f evolving under the Markov transition kernel,

$$= \langle d, Ug \rangle_\mu \quad (60)$$

4 Locally-linear stochastic Perron-Frobenius framework

4.1 Local linear model

For each embedded state $\mathbf{x} \in \mathcal{M} \subset \mathbb{R}^m$ we fit a weighted least-squares linear map

$$A_\theta(\mathbf{x}) = \arg \min_A \sum_{j \in \mathcal{N}_\theta(\mathbf{x})} w_\theta(\mathbf{x}, \mathbf{x}_j) \|\mathbf{x}_{j+1} - A \mathbf{x}_j\|^2,$$

where $\mathcal{N}_\theta(\mathbf{x})$ is the neighbourhood chosen with bandwidth (or kernel scale) θ and $w_\theta(\mathbf{x}, \mathbf{x}_j) = \exp(-\|\mathbf{x} - \mathbf{x}_j\|^2/\theta^2)$. The residuals

$$\mathbf{r}_j = \mathbf{x}_{j+1} - A_\theta(\mathbf{x}) \mathbf{x}_j, \quad j \in \mathcal{N}_\theta(\mathbf{x}),$$

yield an empirical covariance

$$\Sigma_\theta(\mathbf{x}) = \text{Cov}_{j \in \mathcal{N}_\theta(\mathbf{x})} [\mathbf{r}_j].$$

4.2 Stochastic transition kernel

Replacing the Dirac mass of a deterministic map by a Gaussian centred on the local linear prediction gives

$$k_\theta(\mathbf{y} | \mathbf{x}) = \mathcal{N}(\mathbf{y}; A_\theta(\mathbf{x}) \mathbf{x}, \Sigma_\theta(\mathbf{x})),$$

a data-driven transition density that captures both drift and anisotropic noise.

4.3 Stochastic Perron–Frobenius operator

Acting on a probability density ρ ,

$$(P_\theta \rho)(\mathbf{y}) = \int_{\mathcal{M}} k_\theta(\mathbf{y} \mid \mathbf{x}) \rho(\mathbf{x}) d\mathbf{x}.$$

4.4 Likelihood–based objective

Given an embedding Φ_γ with parameters γ and the observed embedded trajectory $\{\mathbf{x}_t\}_{t=1}^T$, the negative log–likelihood is

$$\mathcal{L}(\gamma, \theta) = - \sum_{t=1}^{T-1} \log k_\theta(\mathbf{x}_{t+1} \mid \mathbf{x}_t),$$

and the joint optimisation problem is

$$(\gamma^*, \theta^*) = \arg \min_{\gamma, \theta} \mathcal{L}(\gamma, \theta).$$

4.5 Continuous–time interpretation

For small sampling interval Δt the update

$$\mathbf{x}_{t+\Delta t} = A_\theta(\mathbf{x}_t) \mathbf{x}_t + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, \Sigma_\theta(\mathbf{x}_t)),$$

is a locally linear Itô SDE. Defining $B(\mathbf{x}) = A_\theta(\mathbf{x}) - I$ and $C(\mathbf{x})C(\mathbf{x})^\top = \Sigma_\theta(\mathbf{x})/\Delta t$, the infinitesimal generator acting on a smooth test function f is

$$(\mathcal{L}f)(\mathbf{x}) = \nabla f(\mathbf{x})^\top B(\mathbf{x}) \mathbf{x} + \frac{1}{2} \text{tr}(\Sigma_\theta(\mathbf{x}) \nabla^2 f(\mathbf{x})).$$

This connects the data–driven operator to Koopman and transfer–operator theory in a principled probabilistic way.

4.6 Residual–variance scaling as a curvature diagnostic

Consider a true flow map F on the embedded manifold and its second–order Taylor expansion at a reference point x^* :

$$F(x^* + \delta x) = F(x^*) + J(x^*) \delta x + \frac{1}{2} \delta x^\top H(x^*) \delta x + O(\|\delta x\|^3),$$

where J is the Jacobian and H the rank–3 Hessian tensor. A local *linear* fit removes the J term, leaving the quadratic remainder as the residual

$$r \approx \frac{1}{2} \delta x^\top H(x^*) \delta x.$$

Neighbourhood scaling. For an isotropic kernel of bandwidth θ the typical neighbour offset obeys $\|\delta x\| \sim \theta$, hence

$$\|r\| \sim \theta^2, \quad r r^\top \sim \theta^4.$$

Taking the empirical covariance of residuals gives

$$\Sigma_\theta(x^*) \approx \frac{\theta^4}{4} H(x^*) H(x^*)^\top.$$

Trace-based variance measure. Define

$$\sigma^2(\theta) = \text{trace}[\Sigma_\theta] \approx C \theta^4, \quad C = \frac{1}{4} \text{trace}[H H^\top].$$

Taking logs,

$$\log \sigma^2(\theta) = \log C + 4 \log \theta,$$

so a *log-log* plot of residual variance versus bandwidth exhibits a straight segment of slope 4 whenever quadratic curvature dominates and measurement noise is negligible.

Interpretation of observed slopes.

<i>Slope in log-log plot</i>	<i>Regime indication</i>
≈ 4	Deterministic curvature dominates (Hessian visible).
≈ 0	Flat residual curve: variance from measurement / process noise.
≈ 2	Fit still contaminated by first-order errors or anisotropy.
curved / > 4	Mixing of regimes, higher-order terms, or non-isotropic cloud.

Practical use.

- Plot $\log(\text{trace } \Sigma_\theta)$ versus $\log \theta$ at multiple locations.
- The θ range where the slope stabilises near 4 marks the *curvature-dominated window*.
- Choose an adaptive bandwidth $\theta(x)$ that stays just below this window to minimise artificial diffusion in the stochastic generator, while retaining numerical stability.

References